

HW 9

Townsend Problem 9.9 (5 pts)

The carbon monoxide molecule, CO, absorbs a photon with a frequency of 1.15×10^{11} Hz, making a purely rotational transition from an $l = 0$ to $l = 1$ energy level. What is the internuclear distance for this molecule?

The energy is exactly $\hbar\omega$ for the simple harmonic oscillator here.

The energy from $l = 0$ to $l = 1$

$$\Delta E = \frac{l\hbar^2}{I} = \frac{\hbar^2}{I} \quad (1)$$

Where

$$I = \mu r_0^2, \text{ so } r_0 = \sqrt{\frac{I}{\mu}}$$

We can set these equal and shove

$$I = \frac{\hbar^2}{\Delta E} \quad (2)$$

$$r_0 = \sqrt{\frac{\hbar^2}{\mu \Delta E}} \quad (3)$$

We can assume that μ is on the order of the nuclear mass M_n , but we can easily get μ here anyways

$$\mu = \frac{m_c m_o}{m_c + m_o} \quad (4)$$

This energy difference is

$$\Delta E = h\nu \quad (5)$$

$$\Delta E = 1.15 * 10^{11} * 6.626 * 10^{-34} \quad (6)$$

$$= 7.62 * 10^{-23} J \quad (7)$$

$$(8)$$

The mass of carbon is

$$\frac{12g}{6.02 \cdot 10^{23}} \quad (9)$$

$$= 1.99 \cdot 10^{-26} kg \quad (10)$$

Similarly, the mass of oxygen is

$$2.6 \cdot 10^{-26} kg \quad (11)$$

this gives us

$$\mu = 1.09 \cdot 10^{-26} \quad (12)$$

$$r_0 = \sqrt{\frac{\hbar^2}{\mu \Delta E}} \quad (13)$$

$$= 1.15 \cdot 10^{-10} m \quad (14)$$

$$= 1.15A \quad (15)$$

where A is angstrom, i don't know the latex and have to rush to finish the pset lol.

Townsend Problem 9.13 (5 pts)

A particle is in the orbital angular momentum state $|l, m\rangle$. Evaluate ΔL_x and ΔL_y for this state. Which states satisfy the equality in the uncertainty relation

$$\Delta L_x \Delta L_y \geq \frac{\hbar}{2} |\langle L_z \rangle|$$

Suggestion: One approach is to use $\hat{L}_x = (\hat{L}_+ + \hat{L}_-)/2$, and so on. Another is to take advantage of the symmetry of the expectation values of L_x^2 and L_y^2 in an eigenstate of $\hat{L}_z$.

Note: This could have been a question about general angular momentum—nothing is specific here to orbital.

$$\hat{L}_x = \frac{\hat{L}_+ + \hat{L}_-}{2} \quad (16)$$

$$\hat{L}_y = \frac{\hat{L}_+ - \hat{L}_-}{2i} \quad (17)$$

The expectation value $\langle \hat{L}_x \rangle$ is

$$\langle lm | \hat{L}_x | lm \rangle \quad (18)$$

$$= \frac{1}{2} \langle lm | \hat{L}_+ + \hat{L}_- | lm \rangle \quad (19)$$

$$= \frac{1}{2} \langle lm | \hat{L}_+ | lm \rangle + \frac{1}{2} \langle lm | \hat{L}_- | lm \rangle \quad (20)$$

$$= 0 \quad (21)$$

We can see this because raising the state changes the ket, but not the bra.

The same for $\langle \hat{L}_y \rangle = 0$

1 $\Delta \hat{L}_x$

We can now look at

$$\langle \hat{L}_x^2 \rangle = \langle l, m | \hat{L}_x \hat{L}_x | lm \rangle \quad (22)$$

$$\frac{1}{4} \langle lm | (\hat{L}_+ + \hat{L}_-)(\hat{L}_+ + \hat{L}_-) | l, m \rangle \quad (23)$$

$$\frac{1}{4} \left\langle lm \left| \hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+ + \cancel{\hat{L}_+^2} + \cancel{\hat{L}_-^2} \right| l, m \right\rangle^0 \quad (24)$$

We know that

$$\hat{L}_- |l, m\rangle = \sqrt{l(l+1) - m(m-1)} \hbar |l, m-1\rangle \quad (25)$$

And that

$$\hat{L}_+ |l, m\rangle = \sqrt{l(l+1) - m(m+1)} \hbar |l, m+1\rangle \quad (26)$$

Therefore,

$$\langle lm | \hat{L}_+ \hat{L}_- | lm \rangle \quad (27)$$

$$= \sqrt{l(l+1) - m(m-1)} \hbar \langle lm | \hat{L}_+ | l, m-1 \rangle \quad (28)$$

$$= \sqrt{l(l+1) - m(m-1)} \sqrt{l(l+1) - (m-1)m} \hbar^2 \langle lm | | l, m \rangle \quad (29)$$

$$= (l(l+1) - m(m-1)) \hbar^2 \quad (30)$$

And

$$\langle l, m | \hat{L}_- \hat{L}_+ | lm \rangle \quad (31)$$

$$= \sqrt{l(l+1) - m(m+1)} \hbar \langle l, m | \hat{L}_- | l, m+1 \rangle \quad (32)$$

$$= \sqrt{l(l+1) - m(m+1)} \sqrt{l(l+1) - (m+1)(m)} \hbar^2 \quad (33)$$

$$= (l(l+1) - m(m+1)) \hbar^2 \quad (34)$$

The expectation value is therefore

$$\langle \hat{L}_x^2 \rangle = \hbar^2 \left(\frac{l(l+1) - m(m-1)}{4} + \frac{l(l+1) - m(m+1)}{4} \right) \quad (35)$$

$$(36)$$

The m bits combine

$$m(m-1) - m(m+1) \quad (37)$$

$$m(m-1 - m-1) \quad (38)$$

$$-2m \quad (39)$$

so we get

$$\langle \hat{L}_x^2 \rangle = \frac{2l(l+1) - 2m}{4} \quad (40)$$

This gives us

$$\Delta \hat{L}_x = \frac{2l(l+1) - 2m}{4}$$

2 $\Delta \hat{L}_y$

$$\langle \hat{L}_x^2 \rangle = \langle l, m | \hat{L}_x \hat{L}_x | lm \rangle \quad (41)$$

$$\frac{1}{4i^2} \langle lm | (\hat{L}_+ - \hat{L}_-)(\hat{L}_+ - \hat{L}_-) | l, m \rangle \quad (42)$$

$$- \frac{1}{4} \left\langle lm \left| -\hat{L}_+ \hat{L}_- - \hat{L}_- \hat{L}_+ + \hat{L}_+^2 + \hat{L}_-^2 \right| l, m \right\rangle \quad (43)$$

$$\frac{1}{4} \langle lm | \hat{L}_+ \hat{L}_- + \hat{L}_- \hat{L}_+ | l, m \rangle \quad (44)$$

We can see that this is the same as $\hat{L}_x^2$, so

$$\Delta \hat{L}_y = \frac{2l(l+1) - 2m}{4}$$

Townsend Problem 9.23 (5 pts)

The Hamiltonian for a three-dimensional system with *cylindrical* symmetry is given by

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2\mu} + V(\hat{\rho})$$

where $\rho = \sqrt{x^2 + y^2}$.

- Use symmetry arguments to establish that both $\hat{p}_z$, the generator of translations in the z direction, and $\hat{L}_z$, the generator of rotations about the z axis, commute with $\hat{H}$.
- Use the fact that $\hat{H}$, $\hat{p}_z$, and $\hat{L}_z$ have eigenstates in common to express the position-space eigenfunctions of the Hamiltonian in terms of those of $\hat{p}_z$ and $\hat{L}_z$. *Suggestion:* Follow a strategy similar to the one that we followed in (9.133) for a spherically symmetric potential except that here we are using the eigenfunctions of $\hat{p}_z$ and $\hat{L}_z$ instead of the eigenfunctions of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$.
- What is the radial equation? *Note:* The Laplacian in cylindrical coordinates is given by

$$\nabla^2 \psi = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{\partial^2 \psi}{\partial z^2}$$

3 A

We know that $\hat{L}_z$ commutes with $\hat{p}^2$. The potential only depends on x and y , so any actions on z don't matter which order they are done in. Therefore, the Hamiltonian commutes with $\hat{L}_z$.

4 B

For convenience, I'll label eigenstates of $\hat{H}$ as $|e\rangle$, eigenstates of $\hat{p}^2$ as $|P\rangle$, eigenstates of $\hat{L}_z$ as $|l\rangle$, and eigenstates of $\hat{p}_z$ as $|p\rangle$

We want to find

$$\langle e, l, m | | \rangle \quad (45)$$

Townsend Problem 9.12 (5 pts)

The wave function for a particle is of the form $\psi(\mathbf{r}) = (x + y + z)f(r)$. What are the values that a measurement of $\mathbf{L}^2$ can yield? What values can be obtained by measuring L_z ? What are the probabilities of obtaining these results? *Suggestion:* Express the wave function in spherical coordinates and then in terms of the $Y_{l,m}$'s.

(46)

$$r = \sqrt{x^2 + y^2 + z^2} \quad (47)$$

$$x = r \sin \theta \cos \phi \quad (48)$$

$$y = r \sin \theta \sin \phi \quad (49)$$

$$z = r \cos \theta \quad (50)$$

We can rewrite this in spherical

$$\psi(r) = (x + y + z)f(r) = \sqrt{r}f(r) \quad (51)$$

$$= (\sin \theta \cos \phi + \sin \theta \sin \phi + \cos \theta)rf(r) \quad (52)$$

The $\cos \theta$ part can be represented by

$$Y_{1,0} = \sqrt{\frac{3}{4\pi}} \cos \theta \quad (53)$$

The $\sin \theta \sin \phi$ can be represented with

$$(Y_{11} + Y_{1,-1}) = -\sqrt{\frac{3}{8\pi}} \sin \theta (e^{i\phi} - e^{-i\phi}) \quad (54)$$

$$= -i\sqrt{\frac{3}{4\pi}} \sin \theta \sin \phi \quad (55)$$

the $\sin \theta \cos \phi$ is just

$$(Y_{1,-1} - Y_{11}) = \sqrt{\frac{3}{8\pi}} \sin \theta (e^{i\phi} + e^{-i\phi}) \quad (56)$$

$$= \sqrt{\frac{3}{4\pi}} \sin \theta \cos \phi \quad (57)$$

Adding these just gives

$$\psi(r) = \sqrt{\frac{4\pi}{3}} (Y_{1,-1} - Y_{11} + Y_{1,1} + Y_{1,-1} + Y_{1,0})rf(r) \quad (58)$$

$$= \sqrt{\frac{4\pi}{3}} (2Y_{1,-1} + Y_{1,0})rf(r) \quad (59)$$

$$\hat{L}_z |l, m\rangle = m\hbar |l, m\rangle \quad (60)$$

When we measure $\hat{L}_z$, we can measure either $-\hbar \frac{2}{3}$ of the time, or $0, \frac{1}{3}$ of the time.

Townsend Problem 9.15 (5 pts)

Show that the spherical harmonics $Y_{l,m}$ are eigenfunctions of the parity operator with eigenvalue $(-1)^l$. *Note:* An inversion of coordinates in spherical coordinates is accomplished by $r \rightarrow r, \theta \rightarrow \pi - \theta$, and $\phi \rightarrow \phi + \pi$. Use (9.148). It may be wise to check the specific examples in (9.151) through (9.153) before attempting the general case (9.148).

We can act the parity operator on the state

$$Y_{lm} = \frac{(-1)^l}{2^l l!} \sqrt{\frac{(2l+1)(l+m)!}{4\pi(l-m)!}} e^{im\phi} \frac{1}{\sin^m \theta} \frac{d^{l-m}}{d(\cos \theta)^{l-m}} \sin^{2l} \theta \quad (61)$$

I will ignore all the constants and only look at how the parity operator acts on the coordinate filled portions of Y_{lm} , which I'll call y_{lm} for now

$$Y_{lm} \propto e^{im\phi} \frac{1}{\sin^m \theta} \frac{d^{l-m}}{d(\cos(\theta))^{l-m}} \sin^{2l}(\theta) = y_{lm} \quad (62)$$

$$\hat{\Pi} y_{l,m} = e^{im(\phi+\pi)} \frac{1}{\sin^m(\pi - \theta)} \frac{d}{d(\cos(\pi - \theta))^{l-m}} \sin^{2l}(\pi - \theta) \quad (63)$$

We know that

$$e^{im\pi} = (-1)^m \quad (64)$$

and

$$\sin(\pi - \theta) = -\sin(\theta) \quad (65)$$

$$\cos(\pi - \theta) = -\cos(\theta) \quad (66)$$

$$(67)$$

This becomes

$$(-1)^m \frac{1}{(-1)^m \sin^m(\theta)} \frac{d}{d((-1)^{l-m}(\cos \theta)^{l-m})} \sin^{2l}(\theta) \quad (68)$$

This has just put a constant onto the original state y_{lm} ,

$$\lambda = \frac{(-1)^m}{(-1)^m (-1)^{l-m}} \quad (69)$$

$$\lambda = (-1)^{m-l} \quad (70)$$

This is different from the expected result, but I don't know how to deal with the derivative with respect to $\cos$, and I'm unfortunately low on time before this is due.

Townsend Problem 9.20 (5 pts)

The wave function of a rigid rotator with a Hamiltonian $\hat{H} = \hat{\mathbf{L}}^2/2I$ is given by

$$\langle \theta, \phi | \psi(0) \rangle = \sqrt{\frac{3}{4\pi}} \sin \theta \sin \phi$$

- (a) What is $\langle \theta, \phi | \psi(t) \rangle$? *Suggestion:* Express the wave function in terms of the $Y_{l,m}$'s and the state in terms of $|\ell, m\rangle$ kets.
- (b) What values of L_z will be obtained if a measurement is carried out and with what probability will these values occur?
- (c) What is $\langle L_x \rangle$ for this state? *Suggestion:* Use bra-ket notation and express the operator $\hat{L}_x$ in terms of raising and lowering operators.
- (d) If a measurement of L_x is carried out, what result(s) will be obtained? With what probability? *Suggestion:* If you have worked out Problem 3.15, you can take good advantage of the expressions for the states $|1, m\rangle_x$. Specifically, take advantage of symmetry and

$$|1, 0\rangle_x = \frac{1}{\sqrt{2}}|1, 1\rangle - \frac{1}{\sqrt{2}}|1, -1\rangle$$

5 a

We can break this up into

$$Y_{11} = -\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi} |1, 1\rangle \quad (71)$$

$$Y_{1,-1} = \sqrt{\frac{3}{8\pi}} \sin \theta e^{-i\phi} |1, -1\rangle \quad (72)$$

The sum of these is

$$(Y_{11} + Y_{1,-1}) = -\sqrt{\frac{3}{8\pi}} \sin \theta (e^{i\phi} - e^{-i\phi}) \quad (73)$$

$$= -i\sqrt{\frac{3}{4\pi}} \sin \theta \sin \phi \quad (74)$$

Therefore,

$$\frac{1}{i}(Y_{11} - Y_{1,-1}) = \langle \theta, \phi | | \psi(0) \rangle \quad (75)$$

The spherical harmonics are energy eigenstates of the Hamiltonian and have the same E

$$Y_{l,m}(t) = e^{-i\frac{E}{\hbar}t} Y_{l,m}(0) \quad (76)$$

Therefore, the time evolved ψ has every component multiplied by the same spinning phase

$$\psi(t) = e^{-i\frac{E}{\hbar}t}\psi(0) \quad (77)$$

6 b

$$\hat{L}_z|l, m\rangle = m\hbar|l, m\rangle \quad (78)$$

We have an equal mixture of $Y_{1,1}$ and $Y_{1,-1}$. This gives a 50-50 chance of either $\hbar$ or $-\hbar$ angular momentum.

7 c

$$\hat{J}_x = \frac{\hat{J}_+ + \hat{J}_-}{2} \quad (79)$$

We can apply the operator and do

$$\langle lm | \hat{J}_x | lm \rangle \quad (80)$$

We can't raise the $Y_{1,1}$ state, nor can we lower the $Y_{1,-1}$ state - this just leaves us with what we started - the lowered $Y_{1,1}$ and the raised $Y_{1,-1}$. This is exactly what we started with, but with a factor of $\frac{1}{2}$.

The eigenvalues of $\hat{J}_+$ are $\sqrt{j(j+1) - m(m+1)}\hbar$ and $\hat{J}_-$ are $\sqrt{j(j+1) - m(m-1)}\hbar$.

For the $Y_{1,-1}$ state being raised, this gives

$$\sqrt{1(2) - (-1)(1))} = \sqrt{2+1} = \sqrt{3} \quad (81)$$

For $Y_{1,1}$ being lowered, this is

$$\sqrt{1(2) - (1)(-1))} = \sqrt{3} \quad (82)$$

We get $2\sqrt{3}$ from these alone, but we have the factor of $\frac{1}{2}$.

Therefore, $\hat{L}_x = \frac{1}{2}(2)\hbar = \sqrt{3}\hbar$

8 d

$$|1,0\rangle_x = \frac{1}{\sqrt{2}}|1,1\rangle - \frac{1}{\sqrt{2}}|1,-1\rangle$$